

Proposition I proved that a central force produces equal triangles SAB , SBC , SCD — all equal areas swept in equal times. Proposition II is its converse: now the equal areas are the PREMISE, and centripetal force is the conclusion. About the fixed centre S the body follows the polygonal path $A B C D E$, and we are GIVEN that it describes areas $\propto$ times. This is the observational direction — exactly what Kepler measured for Mars: if you OBSERVE $\triangle SAB = \triangle SBC = \triangle SCD$, must the force point to S ?